

Tugas Besar Kuliah

TF4073-Sistem Kontrol Hibrid Robus

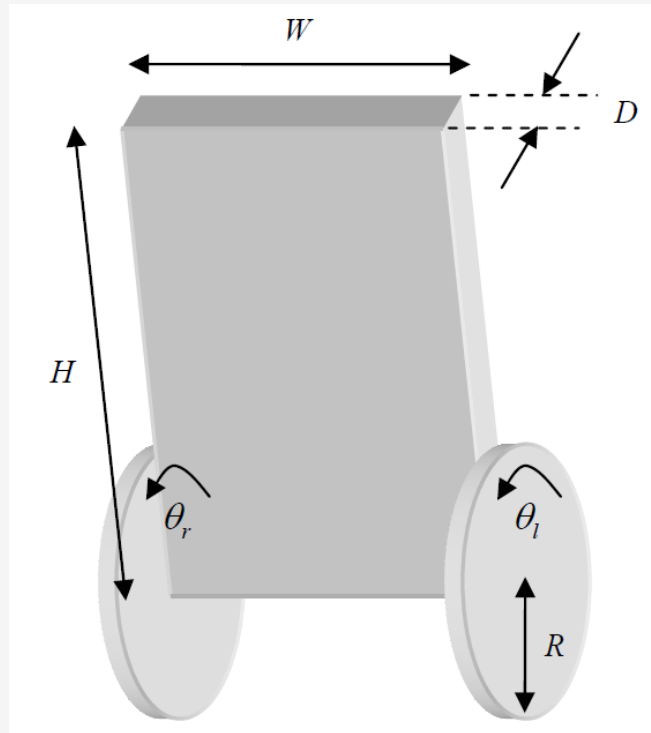
Teja Abdullah Al-Mujahid (13317019)

Fajar Adi Nugroho (13317029)

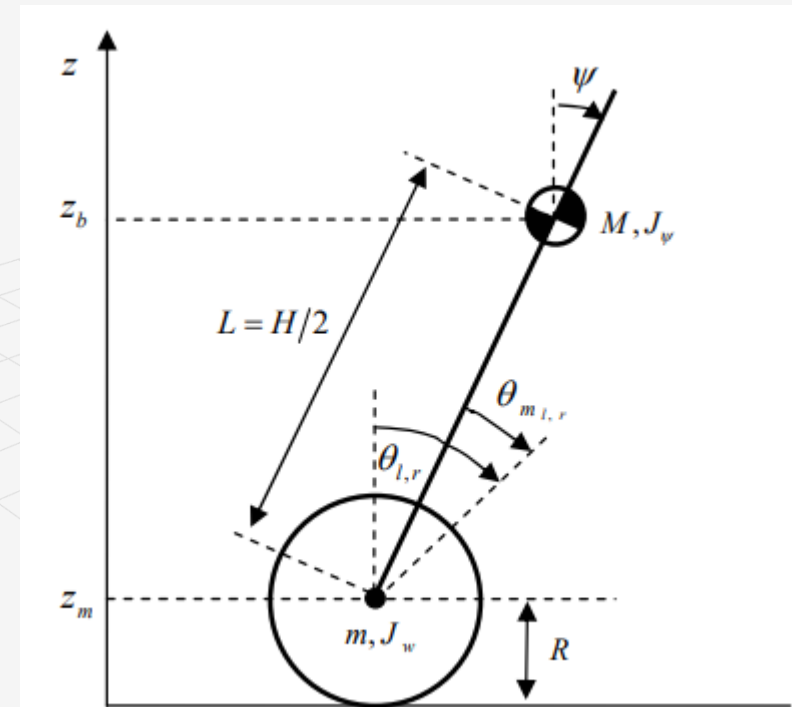
Daniel Beltsazar Marpaung (13317080)

Deskripsi Sistem *Two-Wheeled Inverted Pendulum*

Pemodelan



Sketsa Model Sederhana



ψ : body pitch angle

$\theta_{l,r}$: wheel angle (l, r indicates left and right)

$\theta_{m_{l,r}}$: DC motor angle

Diagram Benda Bebas

Deskripsi Sistem *Two-Wheeled Inverted Pendulum*

Dengan menggunakan metode Lagrangian :

Variabel yang digunakan sebagai koordinat yang digeneralisasi :

- θ : Average angle of left and right wheel
- ψ : Body pitch angle
- ϕ : Body yaw angle

Gaya atau *force* yang ada :

$$F_{\theta} = \frac{\alpha}{2}(v_l + v_r) - (\beta + f_w)\dot{\theta} + \beta\dot{\psi}$$

$$F_{\psi} = -\alpha(v_l + v_r) + 2\beta\dot{\theta} - 2\beta\dot{\psi}$$

$$F_{\phi} = \frac{R}{W}\alpha(v_r - v_l) - \left(\beta + \frac{W}{R}f_w\right)\dot{\phi}$$

$$\alpha = \frac{nK_t}{R_m}, \quad \beta = \frac{nK_t K_b}{R_m} + f_m$$

Persamaan gerak:

$$E \begin{bmatrix} \ddot{\theta} \\ \ddot{\psi} \end{bmatrix} + F \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} + G \begin{bmatrix} \theta \\ \psi \end{bmatrix} = H [v] \quad \leftarrow \text{Tegangan}$$

$$E = \begin{bmatrix} (2m + M)R^2 + 2J_w + 2n^2 J_m & MLR - 2n^2 J_m \\ MLR - 2n^2 J_m & ML^2 + J_{\psi} + 2n^2 J_m \end{bmatrix}$$

$$F = \begin{bmatrix} \beta + f_w & -\beta \\ -2\beta & 2\beta \end{bmatrix}$$

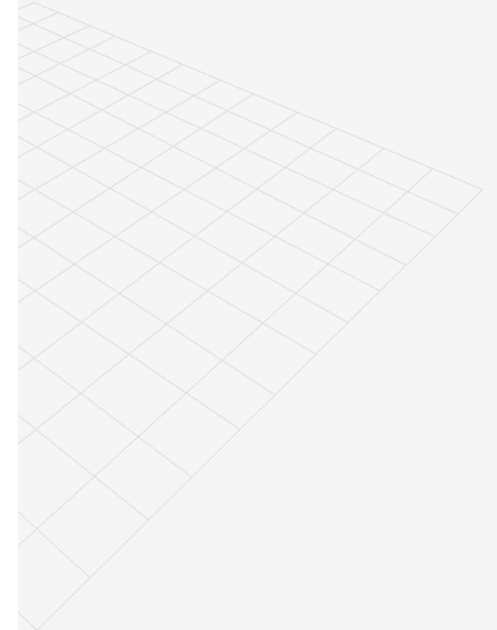
$$G = \begin{bmatrix} 0 & 0 \\ 0 & -MgL \end{bmatrix}$$

$$H = \begin{bmatrix} \alpha/2 & \alpha/2 \\ -\alpha & -\alpha \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Deskripsi Sistem *Two-Wheeled Inverted Pendulum*

Parameter fisik yang digunakan

$g = 9.81$	$[m / \text{sec}^2]$	:	Gravity acceleration
$m = 0.03$	$[kg]$	:	Wheel weight
$R = 0.04$	$[m]$	:	Wheel radius
$J_w = mR^2/2$	$[kgm^2]$	:	Wheel inertia moment
$M = 0.6$	$[kg]$	:	Body weight
$W = 0.14$	$[m]$	:	Body width
$D = 0.04$	$[m]$	:	Body depth
$H = 0.144$	$[m]$	:	Body height
$L = H/2$	$[m]$	:	Distance of the center of mass from the wheel axle
$J_\psi = ML^2/3$	$[kgm^2]$	:	Body pitch inertia moment
$J_\phi = M(W^2 + D^2)/12$	$[kgm^2]$	:	Body yaw inertia moment
$J_m = 1 \times 10^{-5}$	$[kgm^2]$	:	DC motor inertia moment
$R_m = 6.69$	$[\Omega]$	:	DC motor resistance
$K_b = 0.468$	$[V \text{ sec/rad}]$	:	DC motor back EMF constant
$K_t = 0.317$	$[Nm/A]$	:	DC motor torque constant
$n = 1$		:	Gear ratio
$f_m = 0.0022$		:	Friction coefficient between body and DC motor
$f_w = 0$		:	Friction coefficient between wheel and floor.



Deskripsi Sistem *Two-Wheeled Inverted Pendulum*

Persamaan Ruang Keadaan

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$\mathbf{x}_1 = [\theta, \psi, \dot{\theta}, \dot{\psi}]^T \quad u = [v]$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & A_1(3,2) & A_1(3,3) & A_1(3,4) \\ 0 & A_1(4,2) & A_1(4,3) & A_1(4,4) \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 2B_1(3) \\ 2B_1(4) \end{bmatrix}$$

Dan output:

`C_wheel_angle =`

`1 0 0 0`

`C_body_pitch =`

`0 1 0 0`

`C = [C_wheel_angle; C_body_pitch]`

`C =`

`1 0 0 0`
`0 1 0 0`

`D =`

`0`
`0`

Deskripsi Sistem *Two-Wheeled Inverted Pendulum*

Dengan:

$$A_1(3,2) = -MgLE(1,2)/\det(E)$$

$$A_1(4,2) = MgLE(1,1)/\det(E)$$

$$A_1(3,3) = -[(\beta + f_w)E(2,2) + 2\beta E(1,2)]/\det(E)$$

$$A_1(4,3) = [(\beta + f_w)E(1,2) + 2\beta E(1,1)]/\det(E)$$

$$A_1(3,4) = \beta(E(2,2) + 2E(1,2))/\det(E)$$

$$A_1(4,4) = -\beta(E(1,2) + 2E(1,1))/\det(E)$$

$$B_1(3) = \alpha(E(2,2)/2 + E(1,2))/\det(E)$$

$$B_1(4) = -\alpha(E(1,2)/2 + E(1,1))/\det(E)$$

$$\det(E) = E(1,1)E(2,2) - E(1,2)^2$$

State Matrix

A =

0	0	1.0000	0
0	0	0	1.0000
0	-409.7184	-162.5990	162.1273
0	269.6273	78.3429	-78.1496

Input Matrix

B =

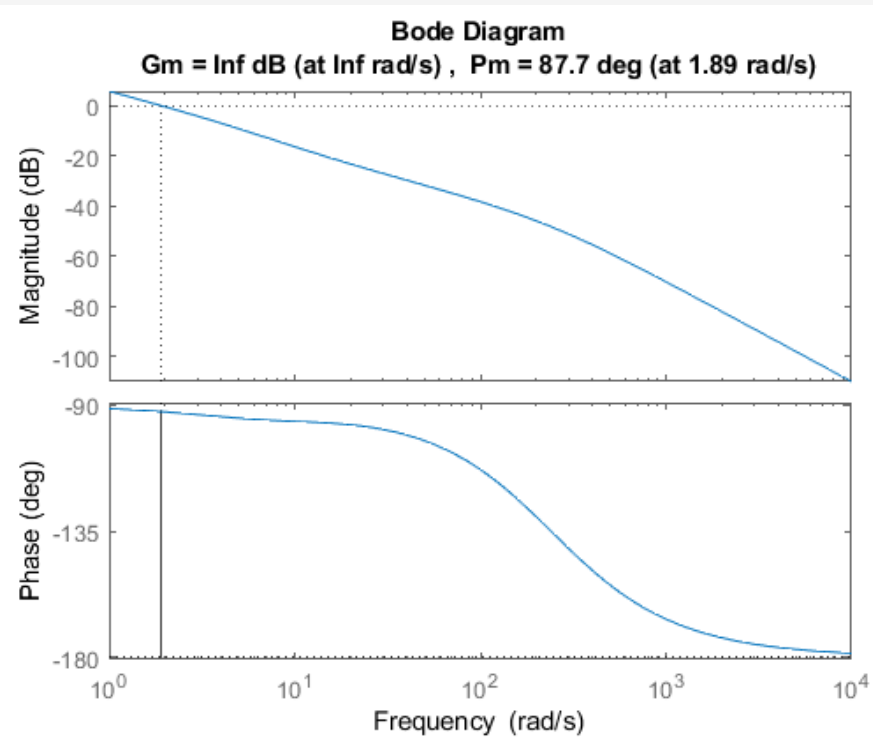
0
0
315.1597
-151.9152

Deskripsi Sistem *Two-Wheeled Inverted Pendulum*

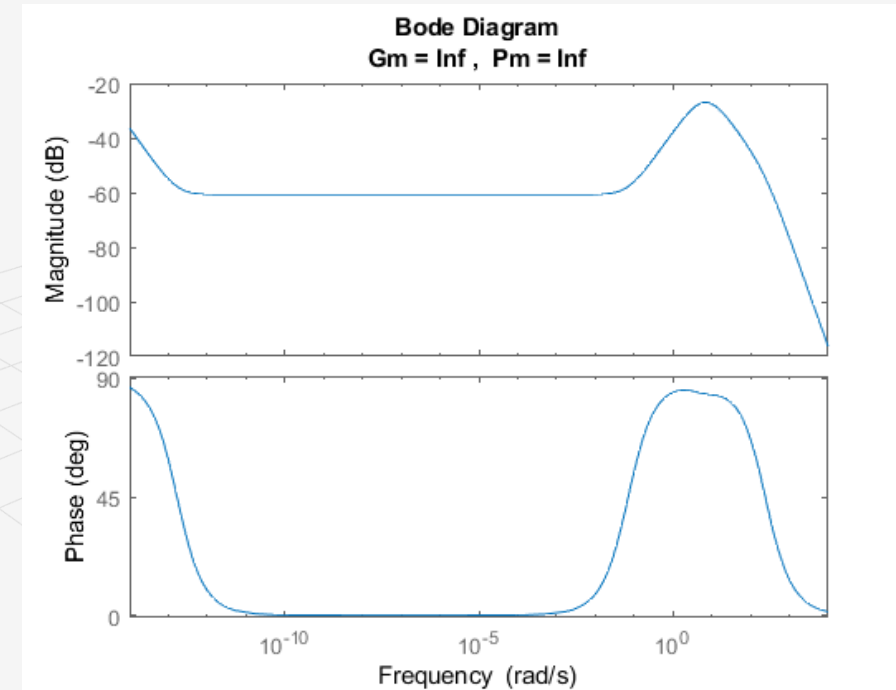
Eigenvalues

```
>> eig_vals'  
  
ans =  
  
    0 -241.6405    7.4312   -6.5393
```

Bode Plot (wheel angle)



Bode Plot (body pitch)



Controllable and Observable

Deskripsi Sistem *Two-Wheeled Inverted Pendulum*

Code

Model Parameter

```
g = 9.81; m = 0.03; R = 0.04;
J_w = m*R^2/2; M = 0.6; W = 0.14;
D = 0.04; H = 0.144; L = H/2;
J_pitch = M*L^2/3; J_yaw = M*(W^2+D^2)/12; J_m = 1.0*10^(-5);
R_m = 6.69; K_b = 0.468; K_t = 0.317;
n = 1; f_m = 0.0022; f_w = 0.0001;

e_11 = (2*m + M)*R^2 + 2*J_w + 2*n^2*J_m;
e_12 = M*L*R - 2*n^2*J_m;
e_22 = M*L^2 + J_pitch + 2*n^2*J_m;
det_E = e_11*e_22 - e_12^2;
alpha = n*K_t/R_m;
beta = n*K_t*K_b/R_m + f_m;
sigma = beta + f_w;
a_32 = -g*M*L*e_12/det_E;
a_42 = g*M*L*e_11/det_E;
a_33 = -2*(sigma*e_22 + beta*e_12)/det_E;
a_43 = 2*(sigma*e_12 + beta*e_11)/det_E;
a_34 = 2*beta*(e_22 + e_12)/det_E;
a_44 = -2*beta*(e_11 + e_12)/det_E;
b_3 = alpha*(e_22 + e_12)/det_E;
b_4 = -alpha*(e_11 + e_12)/det_E;
```

Model Dynamics (State Space)

```
A = [0 0 1 0; 0 0 0 1; 0 a_32 a_33 a_34; 0 a_42 a_43 a_44];
B = 2*[0 0 b_3 b_4]';
C_wheel_angle = [1 0 0 0]; C_body_pitch = [0 1 0 0];
C = [C_wheel_angle; C_body_pitch];
D = [0; 0];

sys = ss(A, B, C, D);

%Eigenvalues
eig_vals = eig(A);
disp('Eigenvalue :');
eig_vals

[b, a] = ss2tf(A, B, C, D);
margin(tf(b(1,:), a));
margin(tf(b(2,:), a));
```

Controllability and Observability

```
Co = ctrb(A, B);
Ob = obsv(A, C);
if rank(Co) == length(A)
    disp('Sistem Controllable'); |
else
    disp('Sistem Tidak Controllable');
end
if rank(Ob) == length(A)
    disp('Sistem Observable');
else
    disp('Sistem Tidak Observable');
end
```

LQR Regulator

Dengan dipilih pembobotan:

$Q =$

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

$R =$

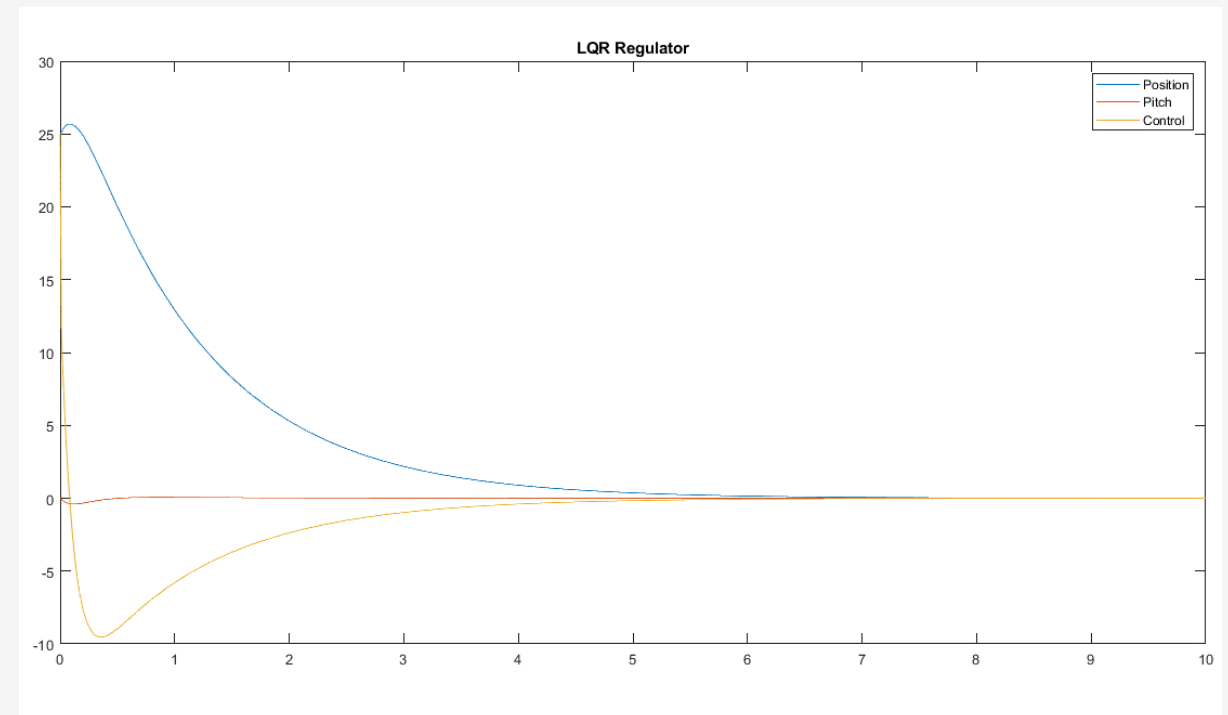
1

Dihasilkan:

$K =$

-1.0000 -48.9587 -1.8729 -5.2054

Dengan kondisi awal $x_0 = [25 \ 0 \ 0 \ 0]^T$, hasil regulasi *output* setelah dikontrol beserta sinyal kontrolnya ditamapkan pada gambar di samping:



*Perpindahan sudut 25 radian ekuivalen dengan posisi 1 m

LQR Regulator

Code

LQR Regulator

```
Q = eye(4);
R = eye(1);
[K, ~, ~] = lqr(A, B, Q, R);

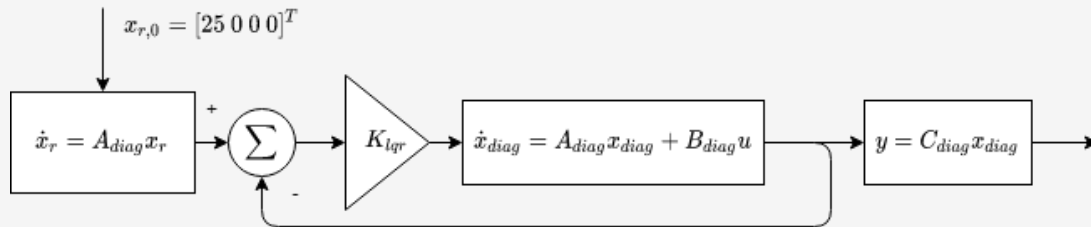
lqr_reg_sys = ss(A-B*K, zeros(4, 1), C, D);
t = 0:0.01:10;
x0 = [25 0 0 0]';
[y, t, x] = initial(lqr_reg_sys, x0, t);
u = -K*x';

disp('LQR Controller Gain');
K

figure;
plot(t, y); hold on;
plot(t, u); legend({'Position', 'Pitch', 'Control'});
title('LQR Regulator');
```

LQR Tracking – Step

Blok Diagram



A_{diag} =

0	0	0	0
0	-241.6405	0	0
0	0	7.4312	0
0	0	0	-6.5393

B_{diag} =

1.9360
348.4994
-2.9015
3.4706

C_{diag} =

1.0000	-0.0037	0.0710	-0.1242
0	0.0018	0.1129	-0.0862

Dengan dipilih pembobotan:

Q =

1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

R =

1

Dihasilkan:

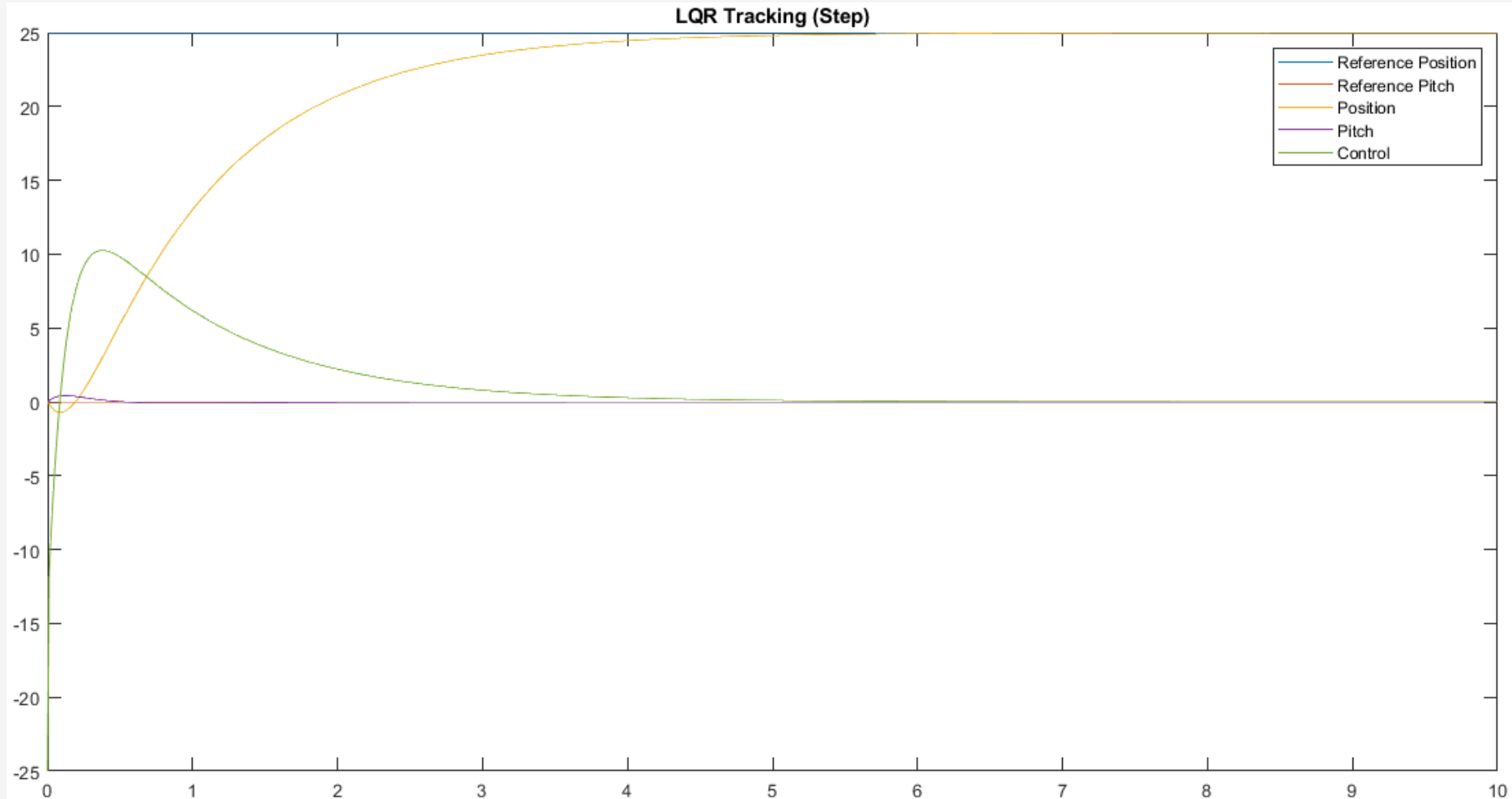
K =

-1.0000	-48.9587	-1.8729	-5.2054
---------	----------	---------	---------

- Pers. ruang keadaan system didiagonlisasi agar mudah mencari *time domain state transition matrix* sehingga $x_{r,0}$ mudah dicari
- $x_{r,0}$ ditentukan sedemikian rupa sehingga $C_{diag} e^{A_{diag} t} x_{r,0} = r(t) = 25$
- Tidak perlu menggunakan *Internal Model Principle* (IMP) karena eigenvalue fungsi *step* (0) telah terdapat pada eigenvalue *plant*

LQR Tracking – *Step*

Hasil *tracking step* posisi, *tracking step 0* (regulasi) *pitch*, dan sinyal kontrolnya



LQR Tracking – Step

Code

LQR Tracking (Step)

```
%Diagonal Canonical Form
[V, J] = eig(A);
A_diag = J;
B_diag = inv(V)*B;
C_diag = C*V;
D_diag = D;

Q = eye(4);
R = eye(1);
[K, ~, ~] = lqr(A_diag, B_diag, Q, R);

%Plant-Reference Augmentation (x_aug = [x_diag x_r]')
A_aug = [A_diag-B_diag*K B_diag*K; zeros(4) A_diag];
B_aug = zeros(8, 2);
C_aug = [zeros(1,4) C_diag(1,:); zeros(1,4) C_diag(2,:); C_diag(1,:) zeros(1,4); C_diag(2,:) zeros(1,4)];
D_aug = [0 0; 0 0; 0 0; 0 0];
lqr_step_sys = ss(A_aug, B_aug, C_aug, D_aug);

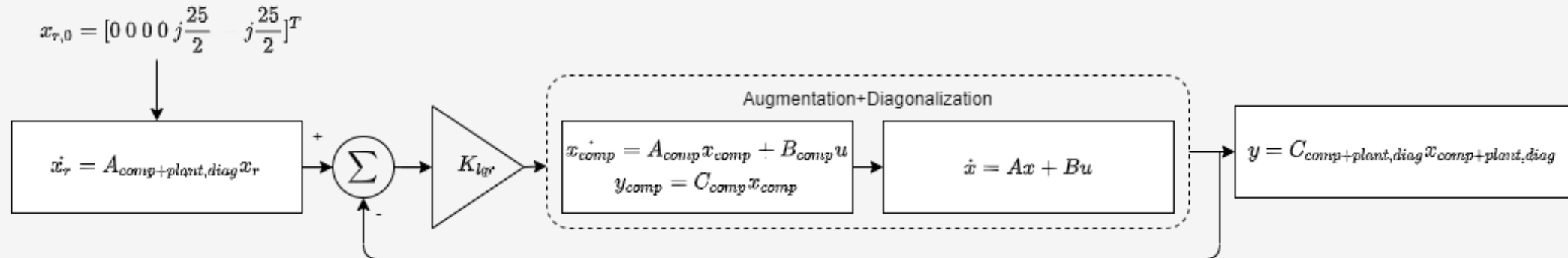
t = 0:0.01:10;
r = [zeros(size(t)) ; zeros(size(t))];
x0 = [zeros(1, 4) 25 0 0 0]';
[y, tout, x] = lsim(lqr_step_sys, r, t, x0);
u = K*(x(:,5:8)-x(:,1:4))';

figure;
plot(tout, y); hold on;
plot(tout, u); legend({'Reference Position', 'Reference Pitch', 'Position', 'Pitch', 'Control'});
title('LQR Tracking (Step)');
```

- Sistem *plant* dan *reference* diaugmentasi sedemikian rupa sehingga $x_{aug} = \begin{bmatrix} x_{diag} & x_r \end{bmatrix}^T$ untuk keperluan realisasi simulasi

LQR Tracking Dengan IMP – Sine Wave

Blok Diagram



Compensator State Space

$A_{cp} =$

$$\begin{bmatrix} 0.0000 + 0.2000i & 0.0000 + 0.0000i \\ 0.0000 + 0.0000i & 0.0000 - 0.2000i \end{bmatrix}$$

$B_{cp} =$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$C_{cp} =$

$$\begin{bmatrix} 0.0000 + 0.5000i & 0.0000 - 0.5000i \end{bmatrix}$$

- Elemen diagonal A_{comp} = eigenvalue sinyal *reference sine* ($j\omega, -j\omega$)
- B_{comp} dan C_{comp} ditentukan sedemikian rupa sehingga $C_{comp} e^{A_{comp} t} B_{comp} = \sin \omega t$

LQR Tracking Dengan IMP – *Sine Wave*

Diagonal Canonical of Plant-Compensator Augmented State Space

A_diag =

1.0e+02 *

Eigen value *plant*

0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0000i	-2.4164 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0000i	0.0000 + 0.0000i	0.0743 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-0.0654 + 0.0000i
0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i

0.0000 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0000i	0.0000 + 0.0000i
0.0000 + 0.0020i	0.0000 + 0.0000i
0.0000 + 0.0000i	0.0000 - 0.0020i

Eigen value *reference*

B_diag =

-9.6798 + 0.0000i
-0.0012 + 0.0000i
0.0105 + 0.0000i
-0.0162 + 0.0000i
5.0347 - 0.0226i
5.0347 + 0.0226i

C_diag =

1.0000 + 0.0000i	-0.0037 + 0.0000i	0.0710 + 0.0000i	-0.1242 + 0.0000i	0.9610 + 0.0000i	0.9610 + 0.0000i
0.0000 + 0.0000i	0.0018 + 0.0000i	0.1129 + 0.0000i	-0.0862 + 0.0000i	-0.0003 + 0.0001i	-0.0003 - 0.0001i

Dengan dipilih pembobotan:

Q =

1	0	0	0	0	0
0	1	0	0	0	0
0	0	1	0	0	0
0	0	0	1	0	0
0	0	0	0	1	0
0	0	0	0	0	1

R =

1

Dihasilkan:

>> K'

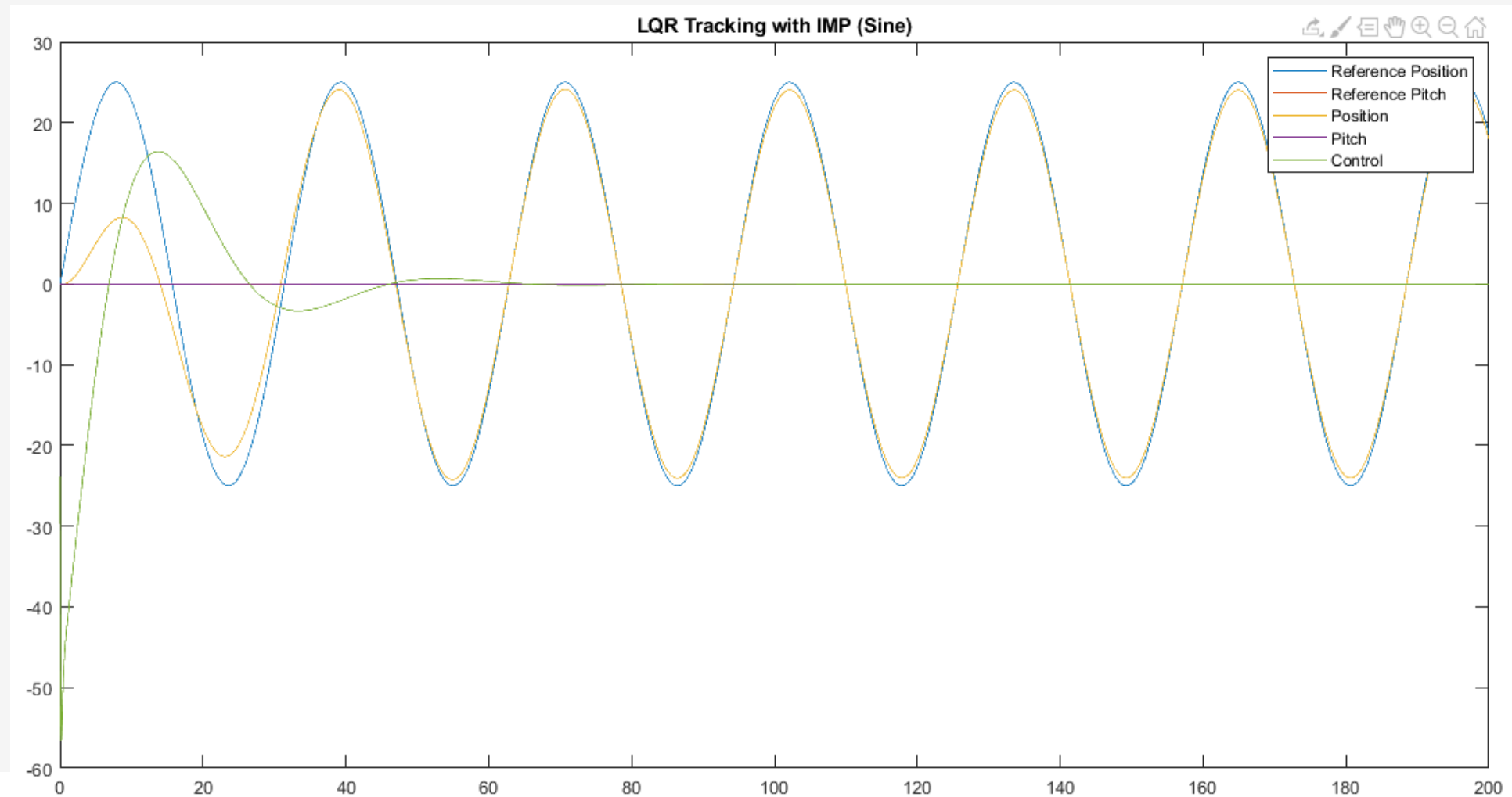
ans =

1.0e+03 *

0.0010 + 0.0000i
-0.0000 + 0.0000i
3.7844 + 0.0000i
0.0000 + 0.0000i
-0.0003 - 0.0010i
-0.0003 + 0.0010i

LQR Tracking Dengan IMP – *Sine Wave*

Hasil *tracking sine* (0.2 rad/s) posisi, *tracking step 0* (regulasi) *pitch*, dan sinyal kontrolnya



LQR Tracking Dengan IMP – *Sine Wave*

Code

LQR Tracking with IMP (Sine)

```
%Compensator
omega = .2;
A_cp = [1i*omega 0; 0 -1i*omega];
B_cp = [1; 1];
C_cp = [1i*1/2 -1i*1/2];
D_cp = [0];

%Plant-Compensator Augmentation (x_aug_1 = [x x_cp]')
A_aug_1 = [A B*C_cp; zeros(2, 4) A_cp];
B_aug_1 = [zeros(4,1); B_cp];
C_aug_1 = [C(1,:) zeros(1,2); C(2,:) zeros(1,2)];
D_aug_1 = [0 0; 0 0];

%Diagonal Canonical Form
[V, J] = eig(A_aug_1);
A_diag = J;
B_diag = inv(V)*B_aug_1;
C_diag = C_aug_1*V;
D_diag = D_aug_1;
```

```
Q = eye(6);
R = eye(1);
[K, ~, ~] = lqr(A_diag, B_diag, Q, R);
disp('LQR Controller Gain');
K

%Plant-Compensator-Reference Augmentation (x_aug_2 = [x_diag x_r]')
A_aug_2 = [A_diag-B_diag*K B_diag*K; zeros(6) A_diag];
B_aug_2 = zeros(12, 2);
C_aug_2 = [zeros(1, 10) 1 1; zeros(1,12); C_diag(1,:) zeros(1,6); C_diag(2,:) zeros(1,6)];
D_aug_2 = [0 0; 0 0; 0 0; 0 0];
lqr_sine_sys = ss(A_aug_2, B_aug_2, C_aug_2, D_aug_2);

t = 0:0.01:200;
r = [zeros(size(t)) ; zeros(size(t))];
x0 = [zeros(1, 6) 0 0 0 0 1i*25/2 -1i*25/2]';
[y, tout, x] = lsim(lqr_sine_sys, r, t, x0);
u = K*(x(:,7:12)-x(:,1:6));

figure;
plot(tout, y); hold on;
plot(tout, u); legend({'Reference Position', 'Reference Pitch', 'Position', 'Pitch', 'Control'});
title('LQR Tracking with IMP (Sine)');
```

- Sistem *plant*, *compensator* dan *reference* diaugmentasi lagi sedemikian rupa sehingga

$$x_{plant+comp+ref,aug} = \begin{bmatrix} x_{plant+comp,diag+aug} & x_r \end{bmatrix}^T \text{ untuk keperluan realisasi simulasi}$$

Kontrol PID-LQR

Matriks A augmentasi :

A_aug =

0	0	1.0000	0	0
0	0	0	1.0000	0
0	-409.7184	-162.5990	162.1273	315.1597
0	269.6273	78.3429	-78.1496	-151.9152
0	0	0	0	0

Matriks B augmentasi :

B_aug =

0
0
0
0
1

Hasil LQR dari matriks augmentasi:

K_aug =

-1.0000	-253.4926	-10.8118	-23.4557	17.6820
---------	-----------	----------	----------	---------

Parameter PID:

Kd =

-1.4839	-3.1948
---------	---------

Kp =

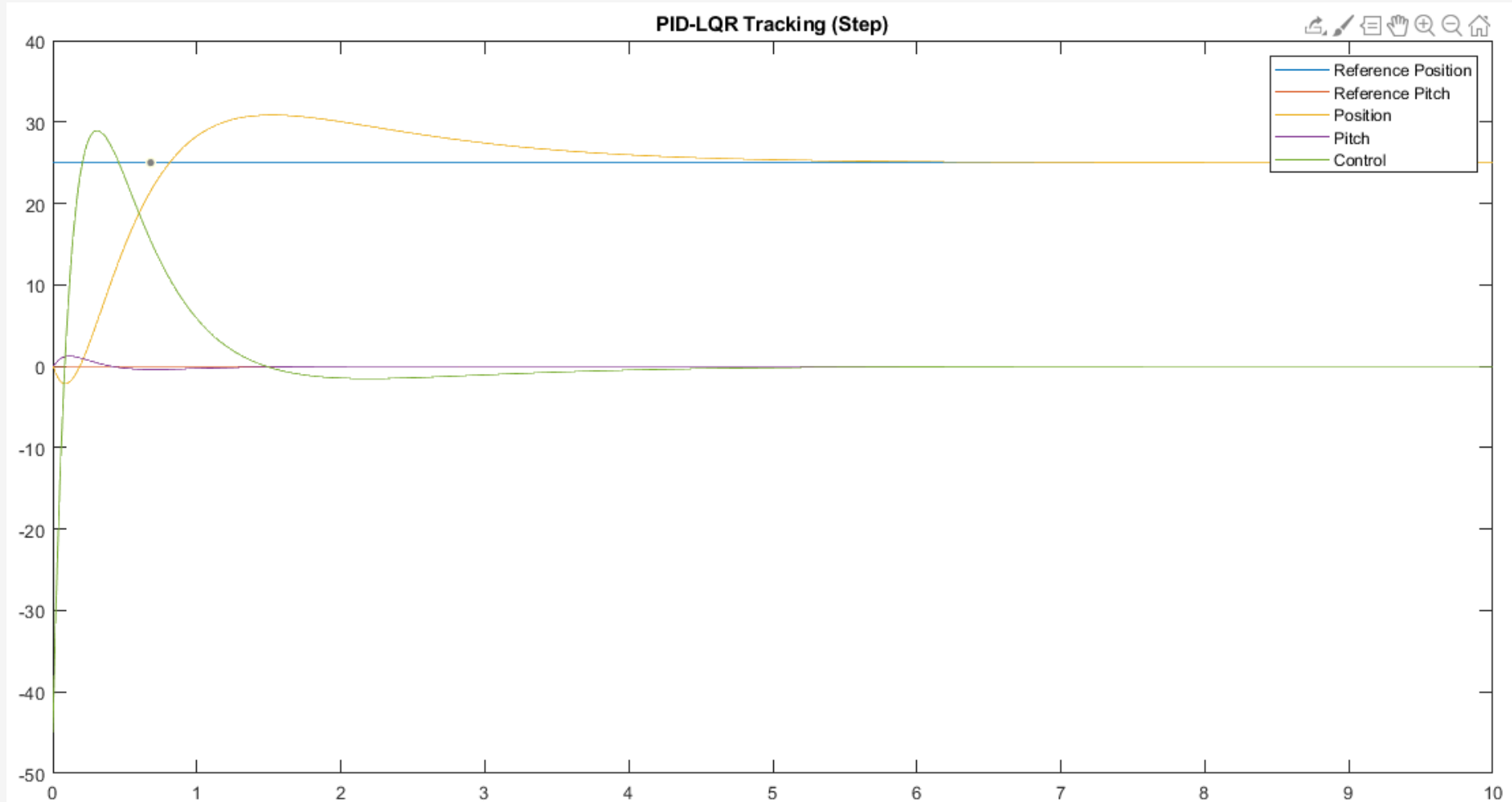
-1.7980	-32.5518
---------	----------

Ki =

-1.0000	-0.0551
---------	---------

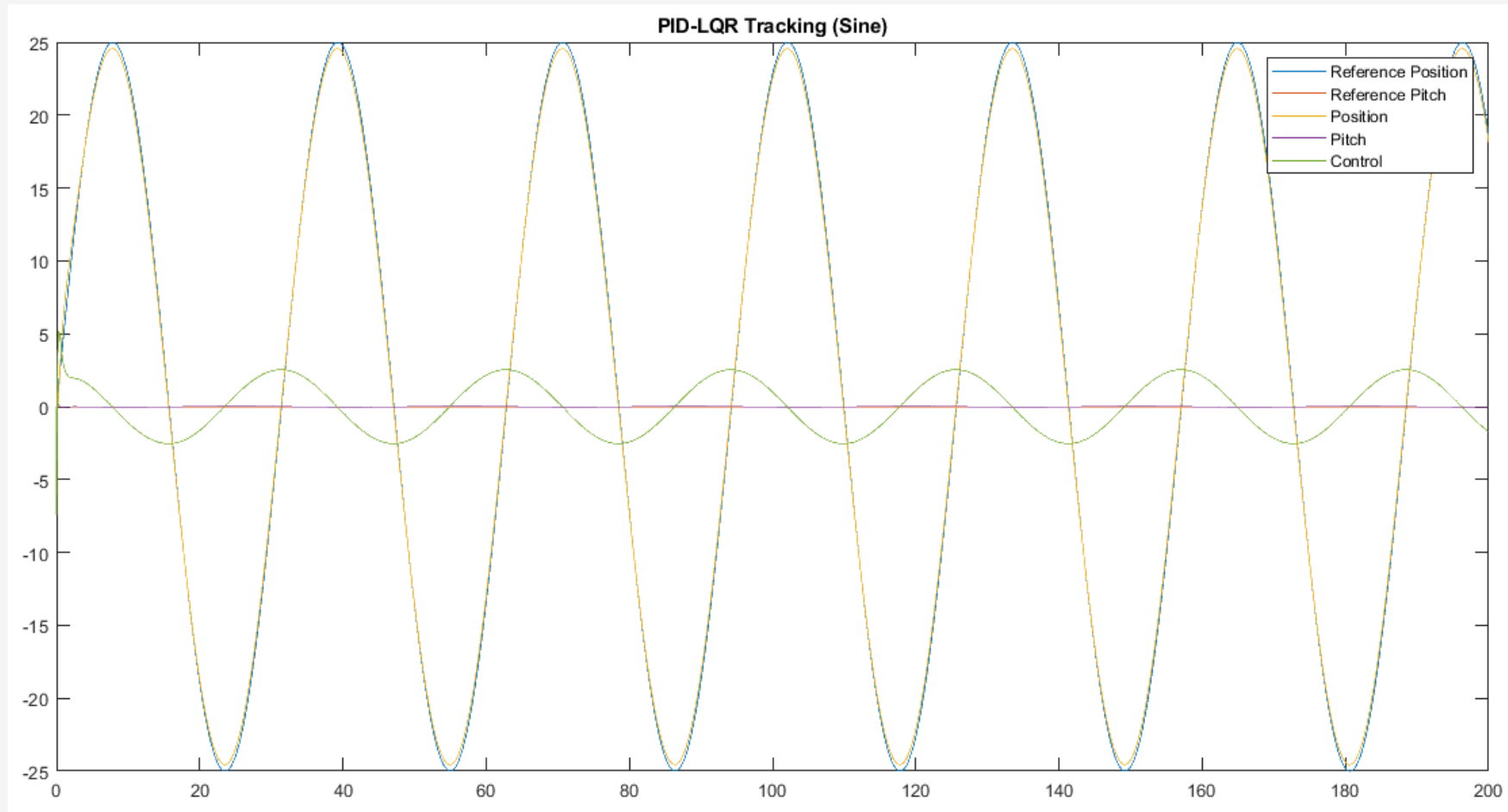
Kontrol PID-LQR

Hasil *tracking step* posisi, *tracking step 0* (regulasi) *pitch*, dan sinyal kontrolnya



Kontrol PID-LQR

Hasil *tracking sine* (0.2 rad/s) posisi, *tracking step 0* (regulasi) *pitch*, dan sinyal kontrolnya



Kontrol PID-LQR

Code

LQR PID

```
%Integral Backstepping Augmentation
A_aug = [A B; zeros(1, 4) 0];
B_aug = [zeros(4, 1); 1];

Q = eye(5);
R = eye(1);
[K_aug, ~, ~] = lqr(A_aug, B_aug, Q, R);

gamma = [C zeros(2, 1); C*A C*B; C*A*A C*A*B];
K_hat = K_aug*pinv(gamma);

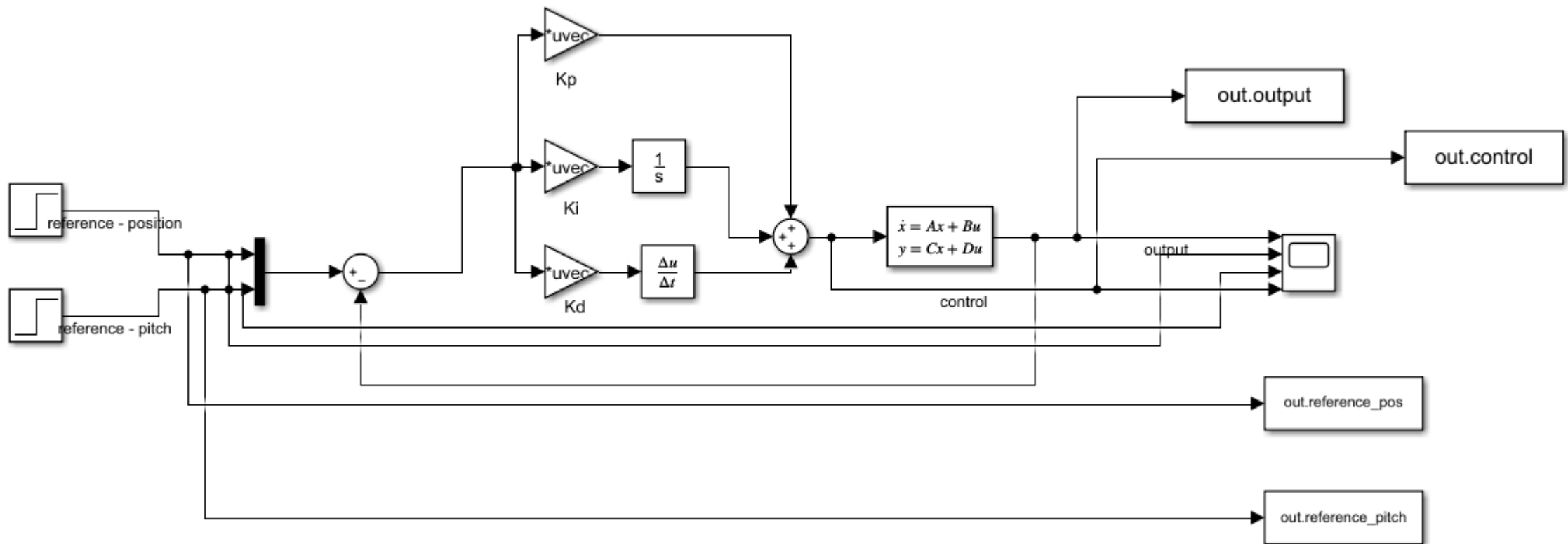
% Parameter PID
K3_hat = K_hat(1,5:6);
Kd = K3_hat/(1+K3_hat*C*B); % Parameter Derivatif
K2_hat = K_hat(1,3:4);
Kp = K2_hat*(1-Kd*C*B); % Parameter Proporsional
K1_hat = K_hat(1,1:2);
Ki = K1_hat*(1-Kd*C*B); % Parameter Integral

simOut = sim('lqr_pid_step');
figure;
plot(simOut.tout, simOut.reference_pos); hold on;
plot(simOut.tout, simOut.reference_pitch); hold on;
plot(simOut.tout, simOut.output); hold on;
plot(simOut.tout, simOut.control); legend({'Reference Position', 'Reference Pitch', 'Position', 'Pitch', 'Control'});
title('PID-LQR Tracking (Step)');

simOut = sim('lqr_pid_sine');
figure;
plot(simOut.tout, simOut.reference_pos); hold on;
plot(simOut.tout, simOut.reference_pitch); hold on;
plot(simOut.tout, simOut.output); hold on;
plot(simOut.tout, simOut.control); legend({'Reference Position', 'Reference Pitch', 'Position', 'Pitch', 'Control'});
title('PID-LQR Tracking (Sine)');
```

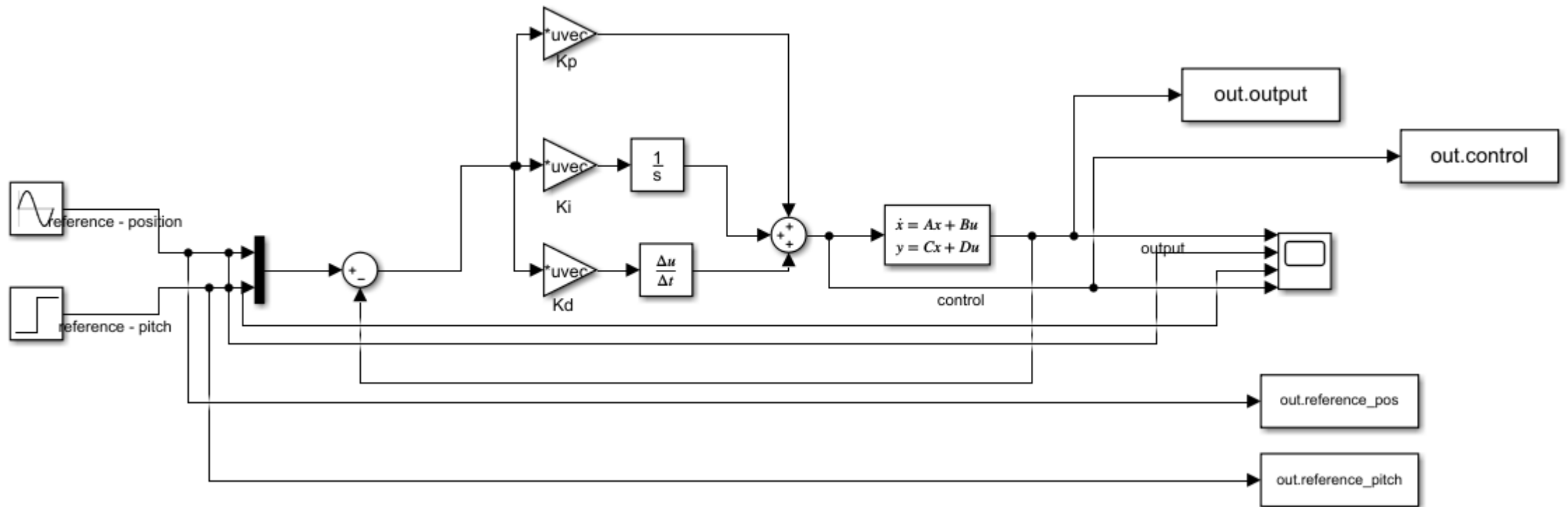
Kontrol PID-LQR

Simulink 'lqr_pid_step'



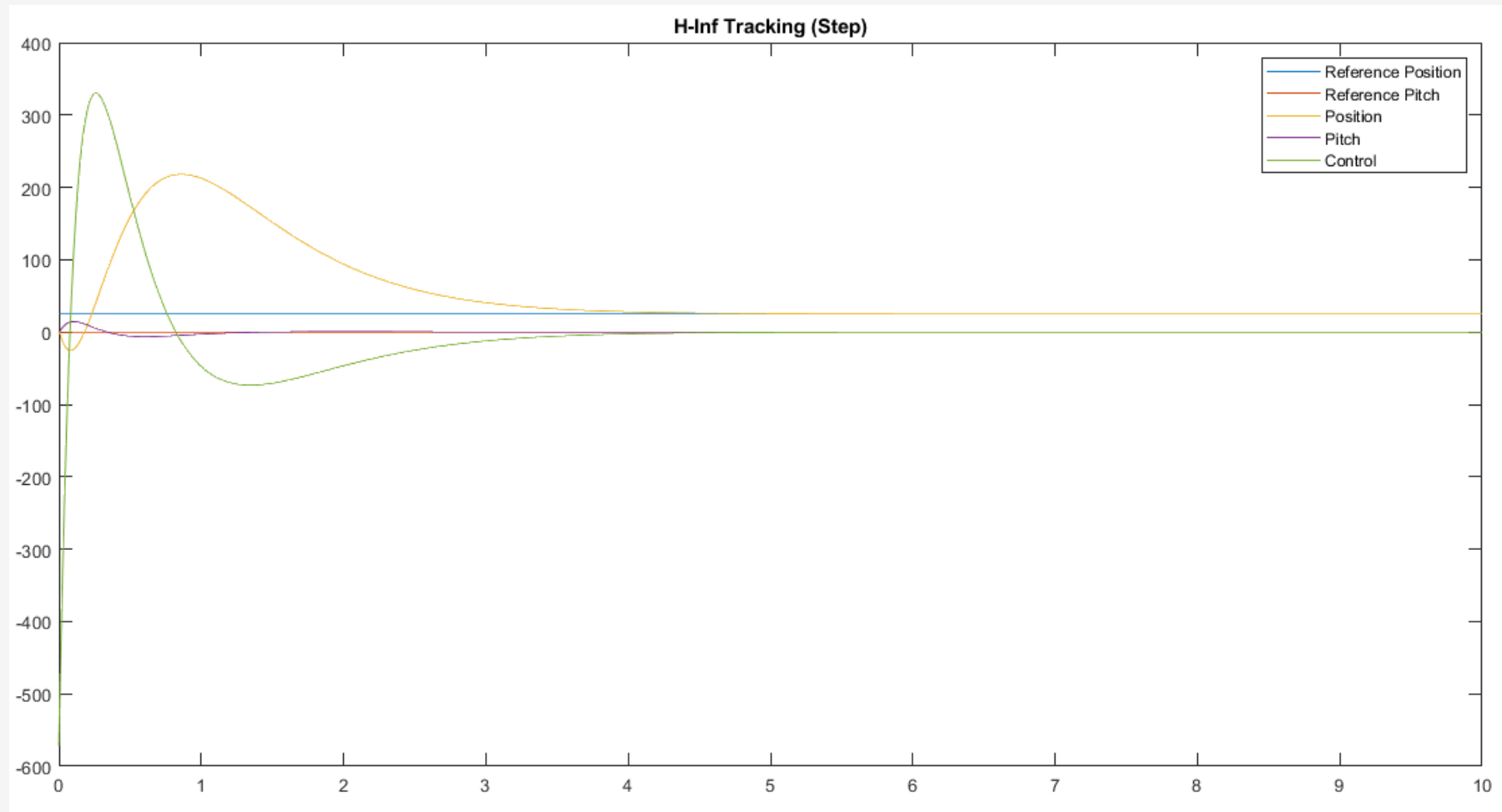
Kontrol PID-LQR

Simulink 'lqr_pid_sine'



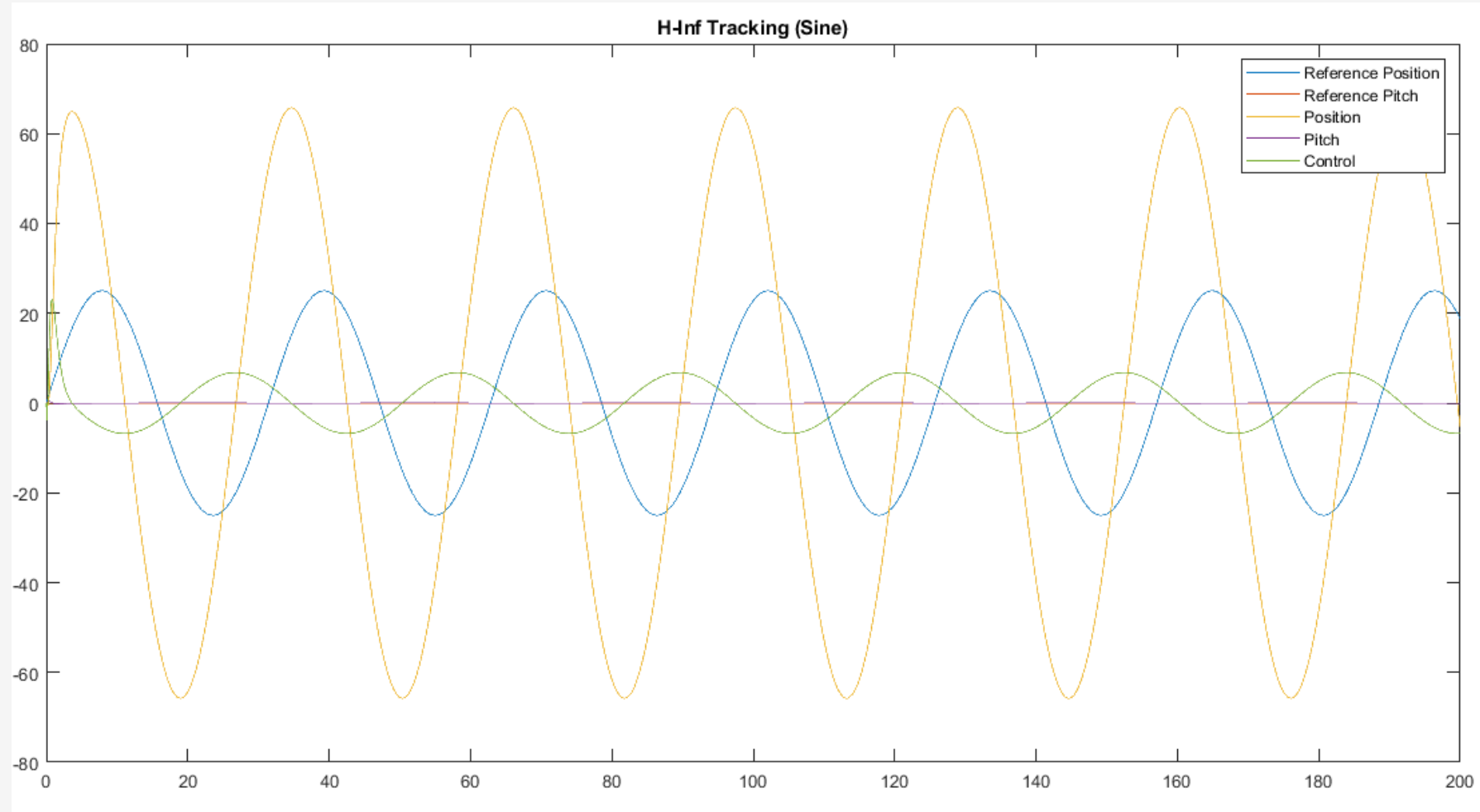
H Infinity (Glover-McFarlane)

Hasil *tracking step* posisi, *tracking step 0* (regulasi) *pitch*, dan sinyal kontrolnya



H Infinity (Glover-McFarlane)

Hasil *tracking sine* (0.2 rad/s) posisi, *tracking step 0* (regulasi) *pitch*, dan sinyal kontrolnya



H Infinity (Glover-McFarlane)

Code

H Inf Glover-McFarlane (ncfsyn) (berhasil)

```
W1 = 1;
W2 = 1;
[K, CL, gamma, info] = ncfsyn(-sys, W1, W2);

disp('Gamma:')|
gamma

hinf_sys = feedback(series(K, sys),eye(2));
hinf_control_sys = feedback(K, sys);

t = 0:0.01:10;
r = 25*[ones(size(t)); zeros(size(t))];
[y, tout, ~] = lsim(hinf_sys,r,t);
[u, tout, ~] = lsim(hinf_control_sys,r,t);

figure;
plot(tout, r); hold on;
plot(tout, y);
plot(tout, u); legend({'Reference Position', 'Reference Pitch', 'Position', 'Pitch', 'Control'});
title('H-Inf Tracking (Step)');

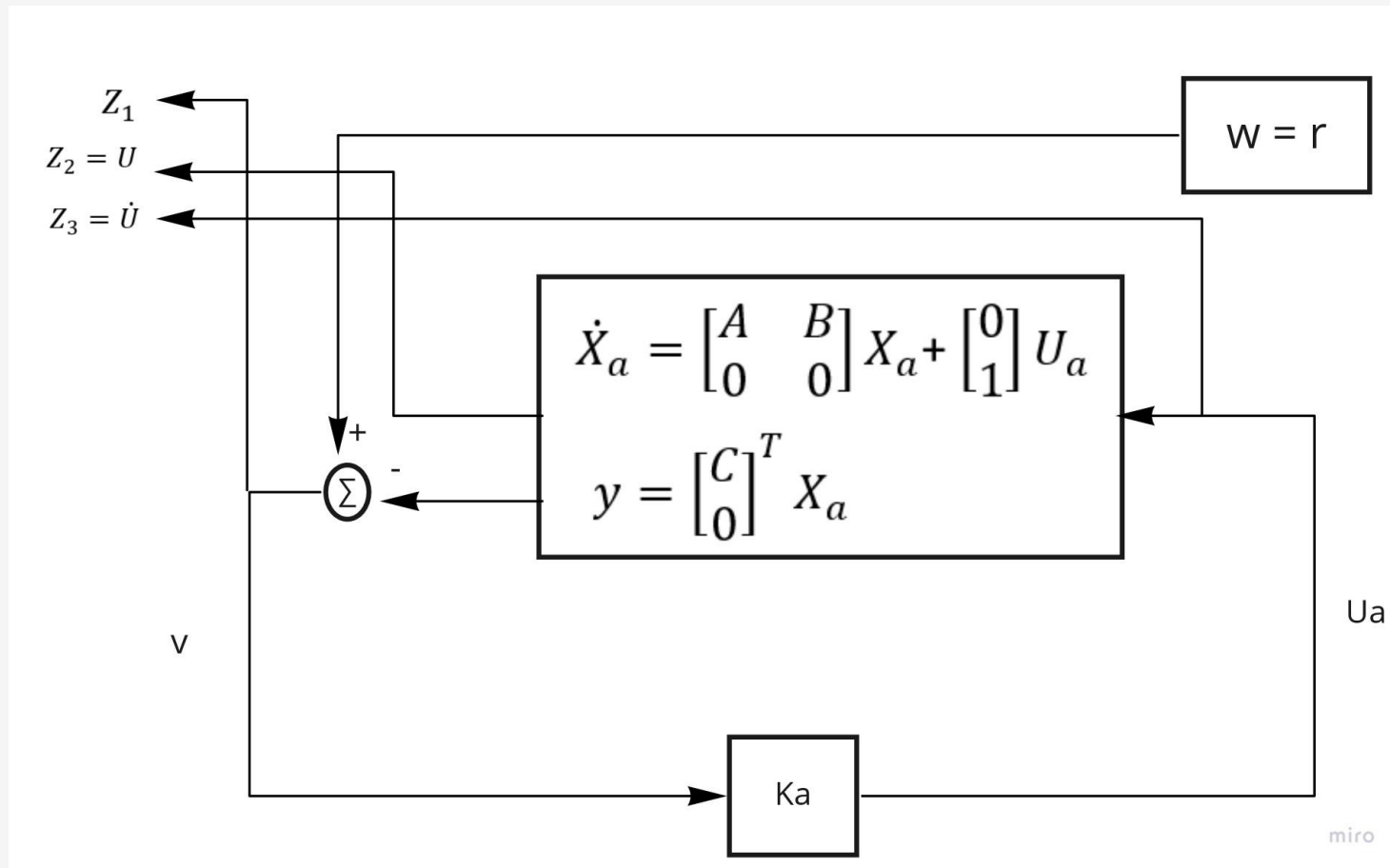
t = 0:0.01:200;
r = 25*[sin(0.2*t); zeros(size(t))];
[y, tout, ~] = lsim(hinf_sys,r,t);
[u, tout, ~] = lsim(hinf_control_sys,r,t);

figure;
plot(tout, r); hold on;
plot(tout, y); hold on;
plot(tout, u); legend({'Reference Position', 'Reference Pitch', 'Position', 'Pitch', 'Control'});
title('H-Inf Tracking (Sine)');
```

- Sebelumnya telah dicoba menggunakan metode Ricatti dan LMI namun menghasilkan *steady state error* yang besar

PID – H Infinity

Diagram Blok



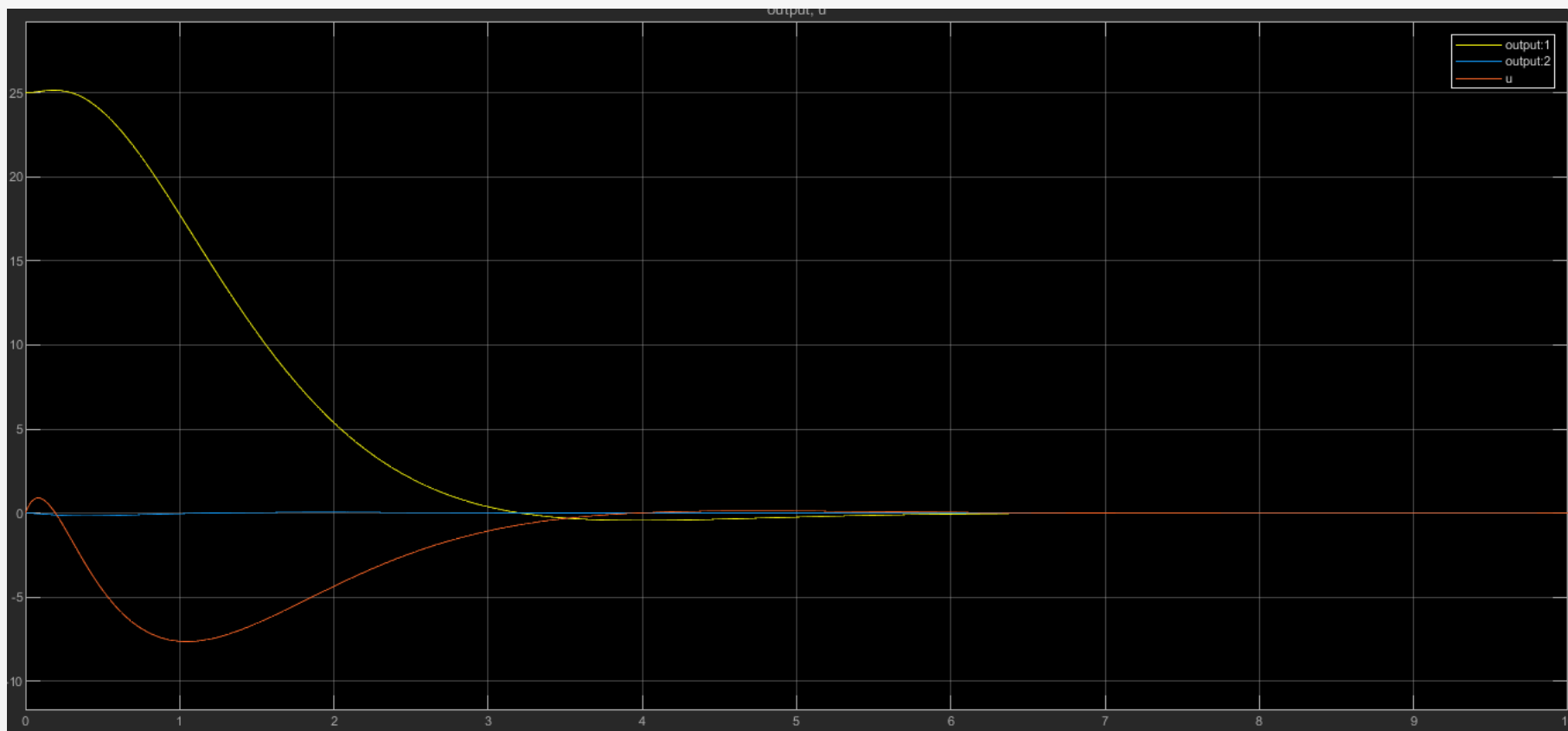
PID – H Infinity

Persamaan matriks

$$\begin{bmatrix} \dot{x} \\ \dot{u} \end{bmatrix} = \underbrace{\begin{bmatrix} A & B_2 \\ 0 & 0 \end{bmatrix}}_{A_a'} \begin{bmatrix} x \\ u \end{bmatrix} + \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_{B_{1a}} \omega + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B_{2a}} u_q$$
$$z = \underbrace{\begin{bmatrix} -C & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}}_{C_a'} \begin{bmatrix} x \\ u \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{D_{1a}} \omega + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{D_{2a}} u_q$$

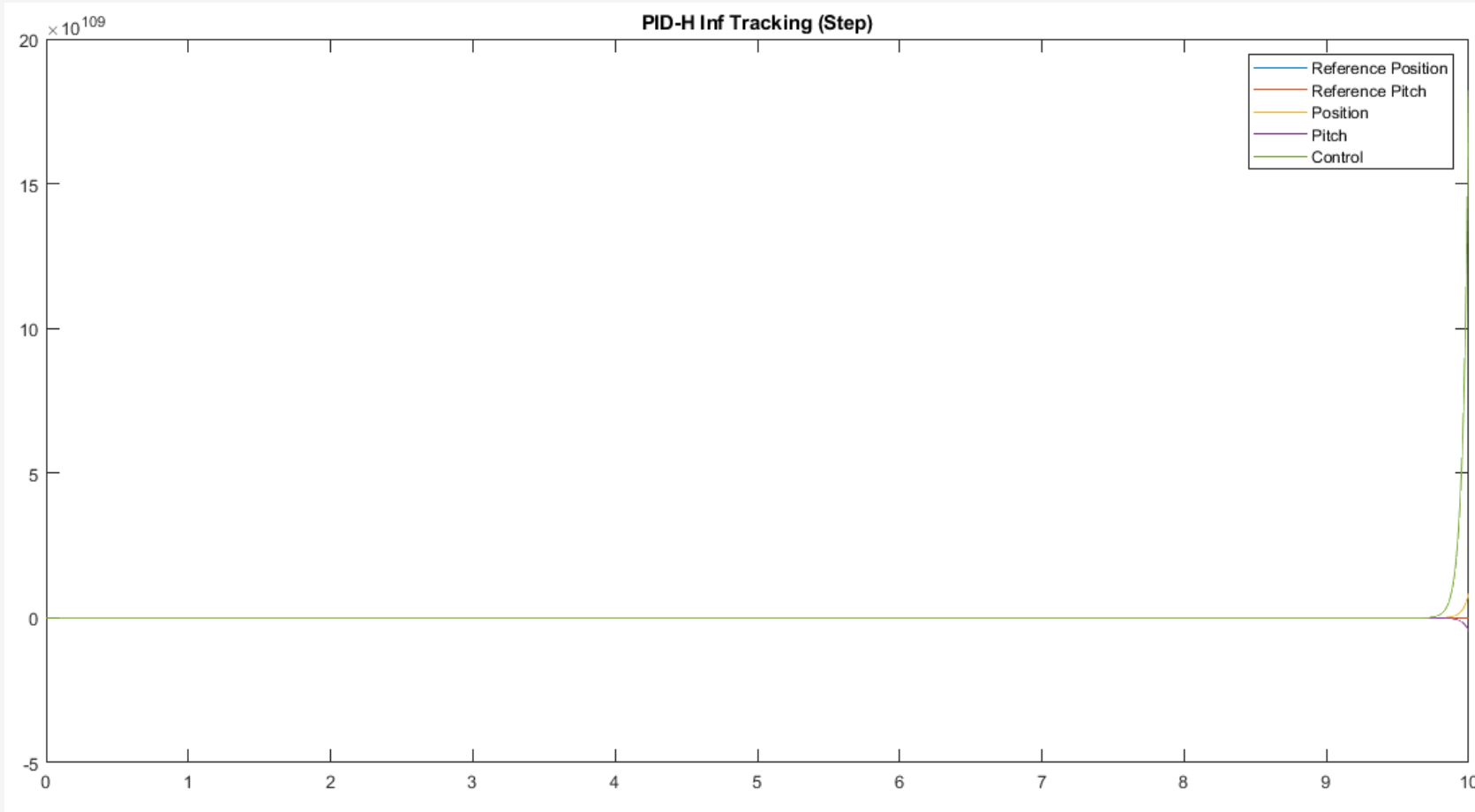
PID – H Infinity

Hasil regulasi *output* dan sinyal kontrolnya untuk bentuk sistem awal (augmentasi, belum diubah menjadi PID)



PID – H Infinity

Hasil *tracking step* posisi, *tracking step 0* (regulasi) *pitch*, dan sinyal kontrolnya untuk bentuk system PID



- Hasil PID tidak stabil (berbeda dengan bentuk augmentasi) menandakan bahwa Moore-Penrose Pseudo Inverse menghasilkan solusi yang kurang tepat

PID – H Infinity

Code

PID - H Inf

```
%Integral Backstepping Augmentation
A_a = A_aug;
B_1a = zeros(5, 1);
B_2a = B_aug;
C_a = [-C zeros(2, 1); zeros(1, 4) 1; zeros(1, 5)];
D_1a = [ones(2, 1); 0; 0];
D_2a = [zeros(3, 1); 1];

gamma2 = 0.1;
A_are = A_a;
B_are = B_2a*B_2a'-gamma2*(B_1a*B_1a');
C_are = C_a'*C_a;
X_are = are(A_are, B_are, C_are);
K_are = -B_aug'*X_are;

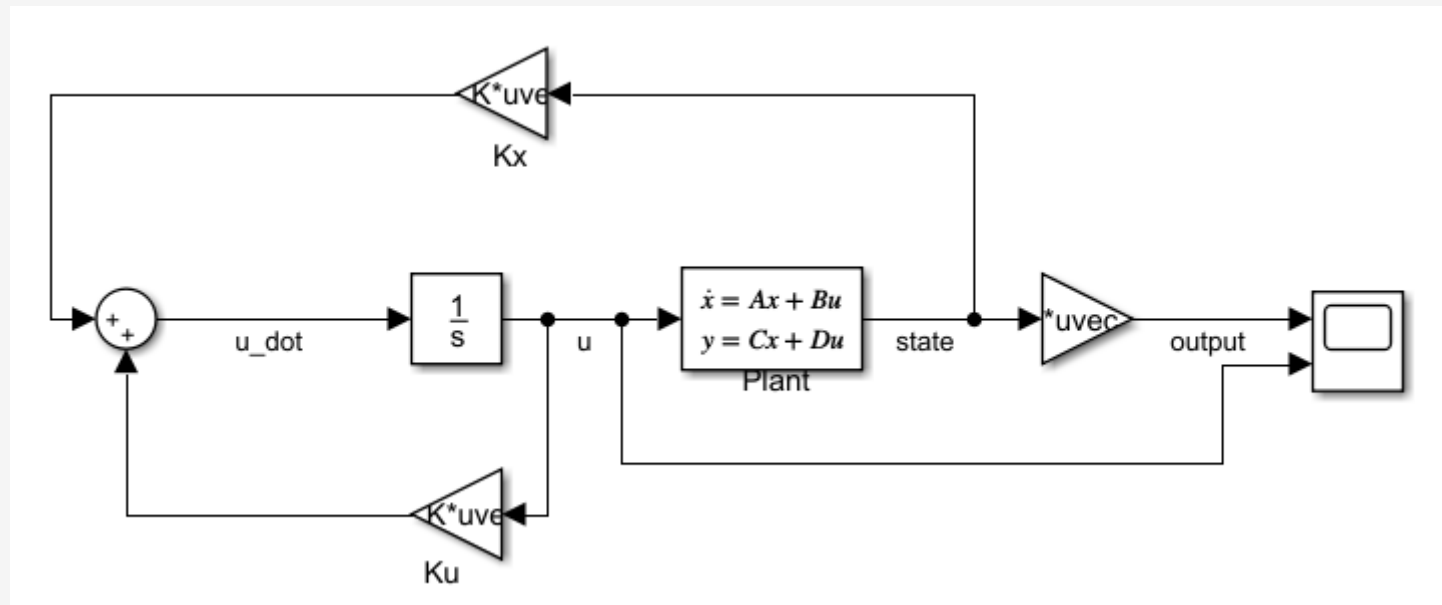
gamma = [C zeros(2, 1); C*A C*B; C*A*A C*A*B];
K_hat = K_are*pinv(gamma);
```

```
% Parameter PID
K3_hat = K_hat(1,5:6);
Kd = K3_hat/(1+K3_hat*C*B); % Parameter Derivatif
K2_hat = K_hat(1,3:4);
Kp = K2_hat*(1-Kd*C*B); % Parameter Proporsional
K1_hat = K_hat(1,1:2);
Ki = K1_hat*(1-Kd*C*B); % Parameter Integral

simOut = sim('lqr_pid_step');
figure;
plot(simOut.tout, simOut.reference_pos); hold on;
plot(simOut.tout, simOut.reference_pitch); hold on;
plot(simOut.tout, simOut.output); hold on;
plot(simOut.tout, simOut.control); legend({'Reference Position', 'Reference Pitch', 'Position', 'Pitch', 'Control'});
title('PID-H Inf Tracking (Step)');
```

PID – H Infinity

Simulink Sistem Augmentasi



References

- Yamamoto, Y. (2008). NXTway-GS Model-Based Design-Control of self-balancing two-wheeled robot built with LEGO Mindstorms NXT. Cybernet Systems Co., Ltd.
- Pradana, W. A., Joelianto, E., Budiyo, A., & Adiprawita, W. (2011). Robust MIMO H_∞ integral-backstepping PID controller for hovering control of unmanned model helicopter. Journal of Aerospace Engineering, 24(4), 454-462.

